

Workshop 1

Requirements Analysis and User Stories

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1. Requirements

Main functional (FR) and non-functional (NFR) requirements of VetCare, a centralized web platform for veterinary clinics. Requirements are grouped by module; quality attributes that apply to the whole system are listed at the end.

Module 1: Authentication & Access Control (RBAC)

- FR_01** The system must allow a new Pet Owner to create a user account by submitting the registration form or activating an invitation issued by the clinic.
- FR_02** The system must allow a new Pet Owner to set the email and password of their account so that they can authenticate later.
- FR_03** The system must allow a registered user to log in by entering their credentials on the login page.
- FR_04** The system must allow a registered user to start an active session when the entered credentials are valid.
- FR_05** The system must allow a user to see an error message when the entered credentials are invalid.
- FR_06** The system must allow an authenticated user to access the pages and endpoints that correspond to their assigned role (Veterinarian, Receptionist, Administrator, Veterinary Assistant, or Owner).
- FR_07** The system must not allow an authenticated user to execute actions outside the permissions of their assigned role.
- FR_08** The system must allow a user to request a time-limited password reset link by entering their registered email.
- FR_09** The system must allow a user to set a new password through a valid reset link.
- FR_10** The system must allow an authenticated user to log out, closing the active session.
- FR_11** The system must end the session of an authenticated user when it expires due to inactivity.
- FR_12** The system must allow an Administrator to create a staff account and assign it a role.
- FR_13** The system must allow an Administrator to edit the details and the role of a staff account.
- FR_14** The system must allow an Administrator to deactivate a staff account so that it can no longer log in.
- FR_15** The system must allow a staff member whose role was changed to use the permissions of the new role.
- NFR_01** The system must store user passwords as salted hashes using a secure algorithm (e.g., bcrypt or Argon2). *[Security]*

- NFR_02** The system must transmit all data between clients and server over TLS (HTTPS).
[Security]
- NFR_03** The system must validate the signature and expiration time of the JWT token on every authenticated request. *[Security]*
- NFR_04** The system must verify the role permissions of the user at the backend layer on every API call. *[Security]*
- NFR_05** The system must temporarily lock an account for 15 minutes after 5 consecutive failed login attempts. *[Security]*
- NFR_06** The system must end a user session after 30 minutes of inactivity. *[Security]*
- NFR_07** The system must validate and sanitize all user input on the server to prevent SQL injection and cross-site scripting (XSS) attacks. *[Security]*

Module 2: Patient & Client Management (CRM & Clinical Records)

- FR_16** The system must allow a Receptionist to save personal details and preferred communication channel of a new customer.
- FR_17** The system must allow a Receptionist to send an invitation to activate the portal account to a newly registered owner.
- FR_18** The system must allow a Receptionist to link a new pet to an existing owner ID to organize multi-pet client accounts.
- FR_19** The system must allow a Veterinarian to open a consultation file with the medical fields specific to the species of the patient (dogs, cats, or exotic animals).
- FR_20** The system must allow a Veterinarian to save the clinical data (vital signs, diagnosis, and treatment) of a consultation in the consultation file.
- FR_21** The system must allow a Veterinarian or Veterinary Assistant to view the complete clinical record (vital signs, diagnoses, treatments, dosages, evolution notes, and surgical history) of a patient.
- FR_22** The system must allow a Pet Owner to view a simplified medical summary (basic info, vaccination status, allergies, and active treatments) of their pet in the health portal.
- FR_23** The system must prevent a Pet Owner from viewing internal veterinary notes in the pet health portal.
- FR_24** The system must allow an authorized user to view the vaccination history, recorded allergies, and previous consultations of a patient.
- FR_25** The system must allow an authorized staff member to search for owners and patients by owner name, national ID, or pet name.
- FR_26** The system must allow a Veterinarian or Veterinary Assistant to record a vaccine application (vaccine, lot number, and application date) in the vaccination history of a patient.

- FR_27** The system must allow a Veterinarian or Veterinary Assistant to view the next due date of a vaccine, calculated automatically from its application date.
- FR_28** The system must allow a Veterinarian to issue a prescription (drug, dose, frequency, and duration) in the consultation record.
- FR_29** The system must allow a Veterinarian to select the inventory item that corresponds to the drug of a prescription.
- NFR_08** The system must return owner and patient search results within 2 seconds under normal load. *[Performance]*
- NFR_09** The system must load the full medical file of a patient within 3 seconds under normal load. *[Performance]*

Module 3: Space & Appointment Management

- FR_30** The system must allow a Pet Owner to submit an online booking request by selecting a pet, a service type, and a time slot, which is registered in pending status.
- FR_31** The system must allow a Receptionist to receive a notification when a Pet Owner submits a new appointment request.
- FR_32** The system must allow a Receptionist to confirm an appointment request.
- FR_33** The system must allow a Receptionist to reschedule an appointment to a new time slot.
- FR_34** The system must allow a Receptionist to cancel an appointment.
- FR_35** The system must allow a Receptionist to see the updated available time slots whenever an appointment is confirmed, rescheduled, or cancelled, to prevent scheduling conflicts.
- FR_36** The system must allow a Receptionist to assign a consultation room to an appointment without double-booking the room or the veterinarian.
- FR_37** The system must allow an Administrator to define the weekly working hours and time off of a veterinarian.
- FR_38** The system must allow a Pet Owner or Receptionist to view the available appointment slots, calculated from the working hours and time off of each veterinarian.
- FR_39** The system must allow a Pet Owner to cancel a pending or confirmed appointment.
- FR_40** The system must allow a Pet Owner to reschedule an appointment to a new available time slot.
- FR_41** The system must allow a Receptionist to receive a notification when a Pet Owner cancels or reschedules an appointment.
- FR_42** The system must allow an authorized staff member to view a visual occupation map of beds and cages with their current availability in the hospitalization module.

- FR_43** The system must allow an authorized staff member to see the status (Critical, Stable, or Under Observation) of each hospitalized patient on the occupation map.
- FR_44** The system must allow a Veterinarian to admit a patient to an available cage.
- FR_45** The system must allow a Veterinarian to set the initial status of a patient when admitting the patient to hospitalization.
- FR_46** The system must allow a Veterinarian to discharge a hospitalized patient, releasing the assigned cage.
- FR_47** The system must allow a Veterinarian to record the discharge date and summary of a hospitalized patient.
- FR_48** The system must allow a Veterinarian or Veterinary Assistant to record the observations of a hospitalization round, together with their staff ID and the timestamp.
- FR_49** The system must allow a Veterinarian to change the status of a hospitalized patient.
- FR_50** The system must allow a Veterinarian to view the log of status changes of a hospitalized patient, with the author and timestamp of each change.
- FR_51** The system must allow a Pet Owner to receive an automated confirmation message via their preferred channel when an appointment is confirmed.
- FR_52** The system must allow a Pet Owner to receive an automated reminder message via their preferred channel when the date of a confirmed appointment approaches.
- FR_53** The system must allow a Pet Owner to receive a vaccination reminder via their preferred channel when the next vaccine due date of their pet approaches.
- NFR_10** The system must update and render the occupation map and the appointment calendar within 2 seconds after a user action. *[Performance]*
- NFR_11** The system must remain operational with 99.5% uptime during regular operational hours. *[Reliability]*
- NFR_12** The system must send automated confirmation and reminder messages within 15 minutes of their scheduled time. *[Performance]*

Module 4: Pharmacy & Inventory Management

- FR_54** The system must allow an authorized staff member to view the available quantities of each pharmaceutical item categorized by batch/lot number and expiration date.
- FR_55** The system must allow an Administrator to register a purchase receipt (items, quantities, lot numbers, expiration dates, and supplier) to add the received items to the inventory.
- FR_56** The system must allow a Veterinarian or Veterinary Assistant to register the drugs prescribed, vaccines applied, and supplies consumed so that the corresponding quantities are deducted automatically from the inventory.

- FR_57** The system must allow an Administrator to receive an automatic alert when a medication or vaccine drops below its minimum threshold or reaches its expiration period.
- FR_58** The system must allow an Administrator to create or edit an inventory item, including its minimum threshold and linked supplier.
- FR_59** The system must allow an Administrator to create or edit a supplier record.
- FR_60** The system must allow an Administrator to view the contact details of the supplier linked to a low-stock medication.
- NFR_13** The system must execute inventory deductions as atomic database transactions to prevent race conditions between concurrent requests. *[Reliability]*

Cross-cutting Non-Functional Requirements

- NFR_14** The system must adapt its layout to the screen size of any device (mobile, tablet, or desktop). *[Usability]*
- NFR_15** The system must work correctly on the latest versions of Chrome, Firefox, Edge, and Safari. *[Compatibility]*
- NFR_16** The system must allow a new staff member to complete the main tasks (register an owner, book an appointment, and record a consultation) after at most one hour of training. *[Usability]*
- NFR_17** The system must meet the WCAG 2.1 level AA guidelines for text contrast and form labeling on its main screens. *[Usability]*
- NFR_18** The system must display error messages in clear language that indicate how to correct the problem. *[Usability]*
- NFR_19** The system must present the interface and notifications in Spanish by default. *[Usability]*
- NFR_20** The system must log the user, action, and timestamp of every view or modification of a clinical record. *[Security]*
- NFR_21** The system must not allow the deletion of clinical entries. *[Reliability]*
- NFR_22** The system must store any correction of a clinical entry as a new version that preserves the previous one. *[Reliability]*
- NFR_23** The system must perform an automated backup of the database every day. *[Reliability]*
- NFR_24** The system must restore normal operation within 4 hours of a failure, with a maximum data loss of 24 hours. *[Reliability]*
- NFR_25** The system must support at least 100 concurrent users without exceeding the response times defined in NFR_10. *[Scalability]*
- NFR_26** The system must allow scaling its components independently. *[Scalability]*

- NFR_27** The system must be built automatically by a CI/CD pipeline (GitHub Actions) on every push. *[Maintainability]*
- NFR_28** The system must be tested automatically by the CI/CD pipeline on every pull request. *[Maintainability]*
- NFR_29** The system must separate presentation, business logic, and data access into distinct layers. *[Maintainability]*
- NFR_30** The system must document its API using the OpenAPI specification. *[Maintainability]*
- NFR_31** The system must record application errors in a log accessible to the development team. *[Maintainability]*

2. User Stories

User stories and acceptance criteria for all relevant roles, written in the format "As a [role], I want to [action] so that [benefit]" with criteria in Given / When / Then form.

Pet Owner / Client

US_OWNER_01: Online Appointment Request

User Story: As a Pet Owner, I want to request an appointment online by selecting my pet, service type, and preferred slot so that I can schedule care without making a phone call.

Acceptance Criteria:

- Given** the owner is logged in and on the appointment scheduling page,
- When** they select their pet, the service type, and an available time slot, and submit the request,
- Then** the system registers the appointment as "Pending Confirmation" and notifies the reception team.

US_OWNER_02: View Simplified Medical Sheet

User Story: As a Pet Owner, I want to view my pet's simplified medical profile and vaccination record so that I can track my pet's health and upcoming vaccinations without confusing technical terms.

Acceptance Criteria:

- Given** the owner is logged into their dashboard,
- When** they navigate to their pet's profile,
- Then** the system displays basic pet details, upcoming vaccine dates, allergies, and last visit summaries, while excluding internal veterinary evolution notes.

US_OWNER_03: Create My Portal Account

User Story: As a Pet Owner, I want to create my account (or activate the invitation sent by the clinic) so that I can see my pets' information and book appointments online.

Acceptance Criteria:

- Given** the owner is on the registration page or has opened an activation link,
- When** they submit a valid email, a password, and their personal details,
- Then** the system creates the account with the Owner role and grants access to the owner portal.

US_OWNER_04: Cancel or Reschedule My Appointment

User Story: As a Pet Owner, I want to cancel or reschedule my appointment online so that the clinic can free the slot and I avoid making a phone call.

Acceptance Criteria:

- Given** the owner has a pending or confirmed appointment,
- When** they choose to cancel or reschedule it and confirm the action,
- Then** the system updates the appointment status, releases the slot, and notifies the reception team.

US_OWNER_05: Receive Appointment and Vaccine Reminders

User Story: As a Pet Owner, I want to receive reminders through my preferred channel so that I do not miss appointments or vaccine doses.

Acceptance Criteria:

- Given** the owner has a confirmed appointment or a vaccine due date approaching,
- When** the reminder date is reached,
- Then** the system sends a message via the owner's preferred channel with the pet name and the appointment date and time or the vaccine name.

Receptionist

US_RECEPTIONIST_01: Owner and Pet Registration

User Story: As a Receptionist, I want to register a new owner and link their pet(s) to their account so that the clinic has accurate contact details and patient associations.

Acceptance Criteria:

- Given** the receptionist is on the registration screen,
- When** they enter the owner's contact information, preferred communication channel, and pet details,
- Then** the system creates the owner profile and links the pet(s) under a single customer record.

US_RECEPTIONIST_02: Manage Appointments & Room Assignment

User Story: As a Receptionist, I want to confirm or reschedule pending appointments and assign an available consultation room so that there are no room or schedule conflicts for veterinarians.

Acceptance Criteria:

- Given** a pending appointment request exists,
- When** the receptionist approves it and assigns an available consultation room and veterinarian,
- Then** the system updates the appointment status to "Confirmed" and ensures no double-booking occurs for that room or doctor.

US_RECEPTIONIST_03: Search Owners and Patients

User Story: As a Receptionist, I want to search owners and pets by name or ID so that I can find their records quickly.

Acceptance Criteria:

- Given** the receptionist is on the client search page,
- When** they enter an owner name, national ID, or pet name,
- Then** the system lists the matching owners with their linked pets and lets the receptionist open the selected record.

Veterinarian

US_VET_01: Species-Specific Medical Record Entry

User Story: As a Veterinarian, I want to create and update a medical record tailored to the patient's species so that I can document vital signs, diagnoses, and treatments accurately.

Acceptance Criteria:

Given a veterinarian is conducting a consultation with a specific animal (e.g., cat or exotic animal),

When they open the medical record form,

Then the system renders input fields customized for that species (e.g., species-relevant temperature ranges and vital sign parameters) and saves the entered clinical diagnosis and prescription.

US_VET_02: Hospitalization Monitoring & Rounds

User Story: As a Veterinarian, I want to view the hospitalization map and record round observations so that I can closely monitor critically ill patients.

Acceptance Criteria:

Given the veterinarian is reviewing hospitalized patients,

When they select a cage on the occupancy map and submit round notes,

Then the system records the timestamped observation note and updates the patient status indicator (e.g., Critical, Stable, Under Observation).

US_VET_03: Vaccinations Record

User Story: As a Veterinarian, I want to register the vaccines applied to a patient so that the vaccination history and the next due dates are always up to date.

Acceptance Criteria:

Given a veterinarian has an open consultation for a patient,

When they record a vaccine with its name, lot number, and application date,

Then the system saves it in the patient's vaccination history, calculates the next due date, and deducts the dose from the inventory.

US_VET_04: Issue a Prescription

User Story: As a Veterinarian, I want to prescribe medication with dose, frequency, and duration so that the treatment is documented and the pharmacy stock is deducted automatically.

Acceptance Criteria:

Given a consultation is open,

When the veterinarian selects a drug from the inventory and enters dose, frequency, and duration,

Then the system saves the prescription in the medical record and deducts the dispensed quantity from stock.

US_VET_05: Admit and Discharge Hospitalized Patients

User Story: As a Veterinarian, I want to admit a patient to an available cage and discharge it when ready so that the occupancy map always reflects reality.

Acceptance Criteria:

- Given** there is at least one free cage,
- When** the veterinarian admits a patient selecting a cage and an initial status,
- Then** the system marks the cage as occupied and shows the patient on the occupancy map.
- Given** a patient is hospitalized,
- When** the veterinarian discharges the patient,
- Then** the system releases the cage and stores the discharge date and summary.

Veterinary Assistant

US_ASSISTANT_01: Record Daily Care & Supplies Consumption

User Story: As a Veterinary Assistant, I want to log daily monitoring notes and supplies consumed during patient care so that patient status is kept up to date and pharmacy stock is accurately maintained.

Acceptance Criteria:

- Given** an assistant completes a routine check or routine care procedure,
- When** they enter daily monitoring values and mark consumed supplies (e.g., syringes, bandages),
- Then** the system saves the monitoring sheet update and automatically deducts the consumed items from inventory.

Clinic Administrator

US_ADMIN_01: Pharmacy Stock Alerts & Supplier Lookup

User Story: As a Clinic Administrator, I want to receive low-stock/expiration alerts and access supplier contact information so that I can reorder necessary pharmaceuticals before running out.

Acceptance Criteria:

- Given** a medication's quantity drops below the minimum defined threshold or is near expiration,
- When** the administrator views the pharmacy dashboard,
- Then** the system highlights the low-stock item with an alert badge and provides direct contact details for the associated supplier.

US_ADMIN_02: Manage Staff Accounts and Roles

User Story: As a Clinic Administrator, I want to create, edit, and deactivate staff accounts and assign roles so that each employee only accesses what their job requires.

Acceptance Criteria:

- Given** the administrator is on the user management page,
- When** they create an account with name, email, and role, or deactivate an existing account,
- Then** the system saves the change, the user's permissions match the assigned role from their next login, and deactivated users can no longer log in.

US_ADMIN_03: Define Veterinarian Schedules

User Story: As a Clinic Administrator, I want to define each veterinarian's working hours and time off so that only valid time slots are offered for appointments.

Acceptance Criteria:

Given the administrator is on the staff schedule page,
When they set a veterinarian's weekly hours and time off and save,
Then the system stores the schedule and offers appointment slots only within the veterinarian's available hours.

US_ADMIN_04: Register Stock Entries and Suppliers

User Story: As a Clinic Administrator, I want to register incoming stock with its lot and expiration date and keep supplier and item records so that the inventory is accurate and organized by batch.

Acceptance Criteria:

Given the administrator is on the inventory management page,
When they register a purchase receipt with items, quantities, lot numbers, expiration dates, and supplier,
Then the system adds the stock, displays it grouped by lot and expiration date, and links each item to its supplier and minimum threshold.

All Users

US_USER_01: Log In Securely

User Story: As a registered user, I want to log in with my credentials so that I can access the features allowed for my role.

Acceptance Criteria:

Given the user is on the login page,
When they enter valid credentials,
Then the system starts a session and shows the home screen for their role.
Given the user is on the login page,
When they enter invalid credentials,
Then the system shows an error message and does not start a session.
Given the user has an active session,
When they log out or remain inactive beyond the session limit,
Then the system invalidates the session and requires a new login.

US_USER_02: Recover My Password

User Story: As a registered user, I want to reset my password through my email so that I can regain access if I forget it.

Acceptance Criteria:

Given the user is on the login page,
When they click "Forgot Password" and enter their registered email,
Then the system sends a time-limited password reset link to that email.

3. User Story Mapping

The map groups the user stories by user activity (columns) and prioritizes them into three releases (rows).

VetCare: User Story Map					
Columns are user activities. Rows are releases in priority order. 20 user stories. FR_03-07 means FR_03 to FR_07.					
	Access & Accounts	Clients & Patients	Appointments	Hospitalization	Pharmacy & Inventory
Release 1: Core clinic operations Accounts, client and patient records, appointment booking with room assignment. 9 stories	US_USER_01 Log in securely All users FR_03-07, FR_10-11 US_OWNER_03 Create portal account Pet Owner FR_01-02 US_ADMIN_02 Manage staff and roles Administrator FR_12-15	US_RECEPTIONIST_01 Register owner and pets Receptionist FR_16-18 US_RECEPTIONIST_03 Search owners and patients Receptionist FR_25 US_VET_01 Species-specific medical record Veterinarian FR_19-21, FR_24	US_ADMIN_03 Define veterinarian schedules Administrator FR_37-38 US_OWNER_01 Request appointment online Pet Owner FR_30-31 US_RECEPTIONIST_02 Confirm and assign room Receptionist FR_32-36		
Release 2: Hospitalization and pharmacy Inpatient care, vaccines, prescriptions and inventory control. 8 stories	US_USER_02 Recover password All users FR_08-09	US_VET_03 Record vaccinations Veterinarian FR_26-27		US_VET_05 Admit and discharge patients Veterinarian FR_44-47 US_VET_02 Monitor map and rounds Veterinarian FR_42-43, FR_48-50 US_ASSISTANT_01 Daily care and supplies Vet. Assistant FR_48, FR_56	US_ADMIN_04 Register stock and suppliers Administrator FR_54-55, FR_58-59 US_VET_04 Issue prescription Veterinarian FR_28-29, FR_56 US_ADMIN_01 Stock alerts and supplier lookup Administrator FR_57, FR_60
Release 3: Owner portal and reminders Owner self-service and automated notifications. 3 stories		US_OWNER_02 View simplified medical sheet Pet Owner FR_22-23	US_OWNER_04 Cancel or reschedule Pet Owner FR_39-41 US_OWNER_05 Receive reminders Pet Owner FR_51-53		
Cross-cutting requirements: NFR_14 to NFR_31 apply to every release. NFR_01 to NFR_07 (security), NFR_20 to NFR_22 (auditability and integrity) and NFR_27 to NFR_28 (CI/CD) are needed from Release 1. Module requirements NFR_08 to NFR_13 are verified in the release that delivers their module.					

4. References

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5. Use of Artificial Intelligence

The team used an AI assistant (Claude, by Anthropic) as a support tool during the preparation of this workshop. Its use covered the following tasks:

- Drafting and improving the wording of requirements and user stories and translating the document into English.
- Reviewing the draft to identify missing requirements and inconsistencies between requirements and user stories and suggesting additions.

The project idea, the scope, the selection of the roles and modules, and the final decision on which requirements, user stories, and releases to include were made by the team members, who reviewed and edited all AI-generated content.